BUSINESS PROBLEM STATEMENT

Title: Iris Flower Classification & Feature Analysis

Background

A botanical research team wants to analyze flower measurement data to:

• Understand structural differences between iris species

• Identify patterns among flower dimensions

• Detect anomalies in measurements

• Build a predictive model for automated species classification

The dataset contains measurements of:

• Sepal Length

• Sepal Width

• Petal Length

• Petal Width

Across three species:

• Setosa

• Versicolor

• Virginica

BUSINESS OBJECTIVES

1. Analyze dataset structure and distribution

2. Identify key distinguishing features

3. Detect unusual observations (outliers)

4. Understand feature relationships

5. Build a classification model for automated prediction

KPI REQUIREMENTS

Total Records

Total number of samples in dataset.

Species Distribution

Count and percentage per species.

Average Feature Values by Species

Mean of each measurement grouped by species.

Correlation Matrix

Understand relationships between features.

Outlier Summary

Count of extreme values using IQR method.

CHART REQUIREMENTS

1. Species Distribution (Pie Chart)

2. Boxplots for Feature Comparison

3. Scatter Matrix (Pair Plot)

4. Correlation Heatmap

MACHINE LEARNING GOAL

Build a classification model that:

• Predicts iris species

• Evaluates accuracy

• Generates performance report